

Diurnal Roost Selection of Northern Long-eared Bats on the Southern Edge of their Range

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Abstract

Mortality from white-nose syndrome has made management and recovery of the northern long-eared bat (*Myotis septentrionalis*) a priority. Ensuring availability of diurnal summer roosts, where gestation and rearing of young occurs, is a critical component of successful recovery. However, observed variation in roost selection across the species' range makes extrapolating data across physiographic regions and habitat types difficult, and limited information exists on roost selection at the southern range extent of the species. Therefore, we examined *M. septentrionalis* diurnal roost selection toward the southern range extent in northern Georgia, USA. We identified characteristics of desirable roosts for all individuals and for reproductive females by comparing roosts to surrounding vegetation and topographic characteristics at the plot, home range (67 ha), and landscape (17.2 km²) scales. From May to July 2014-2017, we radio tagged 12 female and 4 male *M. septentrionalis* and located 37 roosts. Bats roosted in cavities, loose bark, and hollow boles of 16 tree species. Over half of roosts (53%) were located below the canopy and in trees with larger diameters than surrounding trees. Bats disproportionately roosted in snags, with relative probability of selecting a snag roost increasing with snag density at the plot scale. On low gradient slopes, roosts occurred more frequently on west-facing aspects but as slope increased bats showed no aspect preference. Results indicate individuals in our study area are indifferent to or actively avoid roosts with high solar exposure. Our results support previous research that management efforts should focus on creating and retaining snags across the landscape and highlight the need for management to consider regional variation in habitat use.

KEY WORDS Georgia, *Myotis Septentrionalis*, summer, peripheral population, trailing edge

1. Introduction

The arrival of White-Nose Syndrome (WNS) in North America has led to > 90% mortality in several bat species, making long-term population persistence uncertain (Cheng et al. 2021). Compared to pre-WNS populations, surviving little brown bats (*Myotis lucifugus*) have reduced fungal loads and possess variation in immunogenetics, indicating evolutionary rescue may be their best path to recovery in areas where the disease is established (Langwig et al. 2017, Donaldson et al. 2017). Although evolutionary rescue can be a slow process, it can be facilitated by increasing reproductive success in persisting individuals (Kilpatrick 2006, Langwig et al. 2015). For severely impacted bats of North America this will require an understanding summer roosting habitat, which plays a role in the gestation and rearing of young (Kunz 1982).

The northern long-eared myotis's (*Myotis septentrionalis*) mortality from WNS and subsequent federal listing has made conservation of summer roosting habitat a priority (U.S. Fish and Wildlife Service 2016). While *M. septentrionalis* summer roosts are predominantly located in trees, identifying critical summer roosting habitat is difficult due to variation in roost preferences depending on location, sex, forest type, and reproductive status (Broders and Frobes 2004, Carter and Feldhamer 2005, Perry and Thill 2007, Garroway and Broders 2008). Reproductive female bats often select roosts with high solar exposure during pregnancy and rearing of young, which is thought to aid in meeting thermoregulatory requirements (Kalcounis-Rueppell et al. 2005). Results from studies assessing effects of solar exposure on *M. septentrionalis* roosts, regardless of reproductive status, have been inconsistent, with roosts located in areas with high solar exposure and below the canopy where solar exposure is reduced (Menzel et al. 2002, Rojas et al. 2017, Hyzy et al. 2020). Individuals in northern portions of the range select roosts with high solar exposure during energetically expensive life stages, such as

lactation (Garroway and Broders 2008). However, Silvis et al. (2012) found female *M. septentrionalis* in the central portion of the range consistently roosted below the canopy where solar radiation is reduced, regardless of reproductive status. Further variation in roost selection occurs in tree species preference, with some studies showing exclusive use of hardwood trees (Foster and Kurta 1999, Menzel et al. 2002, Carter and Feldhamer 2005) whereas others demonstrate strong preference for conifers (Perry and Thill 2007). Roost tree size selected by female bats also varies, with either no preference for tree size (Menzel et al. 2002, Carter and Feldhamer 2005) or selection for larger diameter trees (Foster and Kurta 1999, Silvis et al. 2015a, Hyzy et al. 2020).

Consistent across *M. septentrionalis* roost studies is use of bark, cavities, and crevices in snags and live trees, with bats displaying a preference for snags (Lacki and Schwierjohann 2001, Lacki et al. 2009, Silvis et al. 2015a). Additionally, reproductive female *M. septentrionalis* are reported to use a roost network (Johnson et al. 2009, Silvis et al. 2015b). Female networks contain 4-42 roosts and cover up to 58.3 ha (Silvis et al. 2015b). Roost switching is frequent within the network, with mean days spent in roosts ranging from 1.3 to 5.3 and most bats switching roosts every 2-3 days (Menzel et al. 2002, Johnson et al. 2009). Movement between roosts typically is ≤ 0.2 km (Carter and Feldhamer 2005, Johnson et al. 2009, Silvis et al. 2015b), but mean distances up to 0.6 km and individual movements up to 2.0 km have been reported (Foster and Kurta 1999, Cryan et al. 2001). *Myotis septentrionalis* use of roost networks across its range indicates that landscape-scale factors could be involved in roost habitat selection. However, studies addressing the influence of landscape factors on *M. septentrionalis* roost selection are limited, but have revealed correlations between roost habitat and fragmented forest (Pauli et al. 2015).

90 The latitudinal range encompassed by *M. septentrionalis* results in a wide temperature
91 gradient. Importance of ambient temperature and optimal thermoregulatory strategies have been
92 demonstrated, with *M. septentrionalis* selecting for increased canopy cover in warmer years
93 (Silvis et al. 2015b). A closely related species, the Indiana bat (*Myotis sodalis*), displays
94 variation in roost selection at its northern and southern range limits. At the northern range extent,
95 individuals select only roosts with direct solar exposure (Kurta et al. 1993), whereas individuals
96 at the southern extent select roosts with high solar exposure as well as roosts with low exposure,
97 such as below the canopy or in areas of high canopy cover (Britzke et al. 2003). Similar variation
98 in *M. septentrionalis* roost selection likely exists at the thermal extremes of the range and
99 highlights the need to understand variation in roost selection at range limits.

100 Few published data are available on roosting habitat of *M. septentrionalis* in the
101 southeastern portion of its range, with most studies occurring in the range core. Given observed
102 variation in roost selection across the range, results from previous roost studies may not apply at
103 the southern periphery. Our goal was to identify characteristics of *M. septentrionalis* roosts
104 toward the southern range limit to inform land managers in these areas tasked with protecting
105 and promoting habitat under current climate conditions. We characterized roost location relative
106 to the surrounding environment to determine factors that affect roost selection at the plot, home
107 range, and landscape scale. Given the importance of thermoregulation during reproduction and
108 our study area being toward the southern periphery of *M. septentrionalis* range where the climate
109 is relatively warmer, we based hypothesis around solar exposure (Kalcounis-Rueppell et al.
110 2005, Garroway and Broders 2008). We hypothesized both bats would select for features
111 associated with reduced solar radiation in an effort to optimize thermoregulation in warmer
112 regions of their range (Kurta et al. 1993, Britzke et al. 2003, Silvis et al. 2015b). Additionally,

we hypothesized that on the landscape scale *M. septentrionalis* may be selecting for forest types that contain a particular mix of trees they prefer as roost or comprised quality forging habitat (Henderson and Broders 2008, Pauli et al. 2015).

2. Methods

2.1 Study Area

Our study was located in northern Georgia, USA, which is toward the southern extent of the historic range of *M. septentrionalis* (Fig.1). The study area includes the Blue Ridge, Valley and Ridge, Appalachian Plateau, and northern portion of the Piedmont physiographic provinces, and ranges topographically from mountainous in the Blue Ridge to rolling hills in the Piedmont (Hodgkins 1965). Forests in these regions are composed of mesophytic and southern mixed forest, with both forest types containing oak (*Quercus*) dominated sections (Dyer 2006). Common tree species include loblolly (*Pinus taeda*), shortleaf (*P. echinata*), white (*P. strobus*), and Virginia (*P. virginiana*) pine, white (*Q. alba*), northern red (*Q. rubra*), and chestnut (*Q. prinus*) oak, yellow poplar (*Liriodendron tulipifera*), and red maple (*Acer rubrum*).

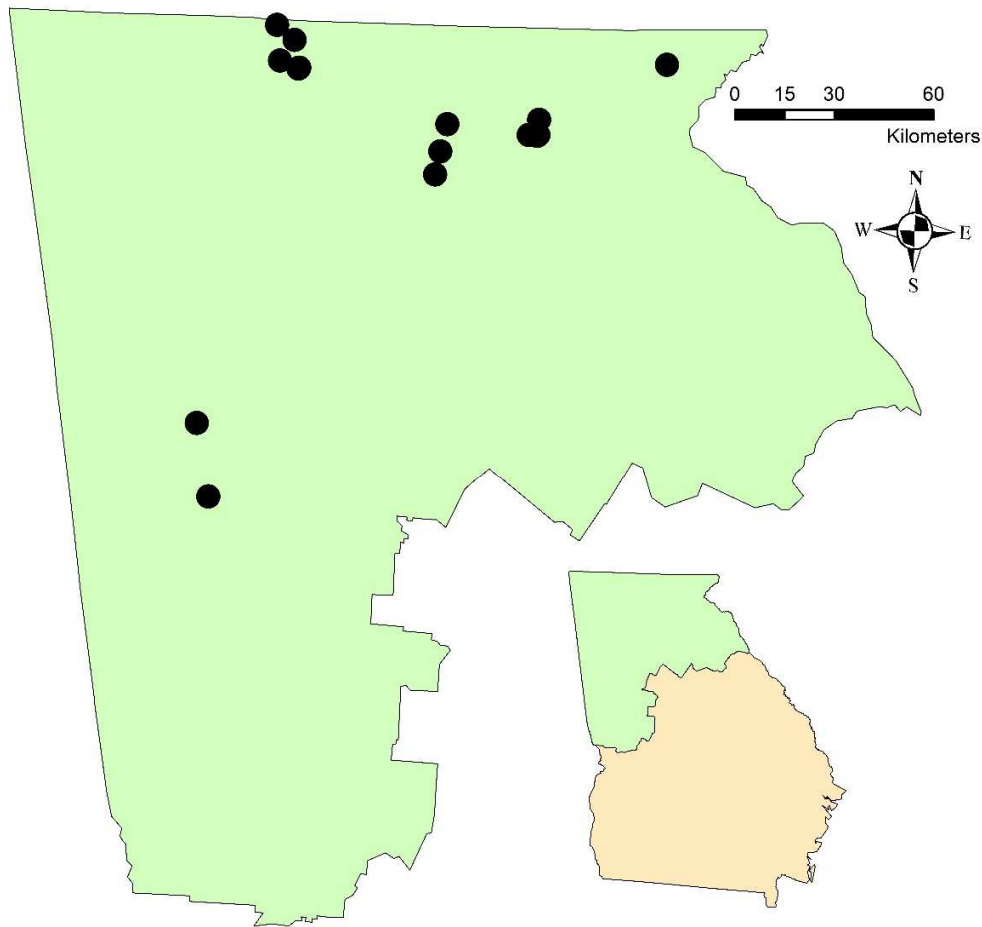


Figure 1. Locations (black dots) of roost centers used to characterize *Myotis septentrionalis* roosts at the landscape scale in north Georgia, USA, 2014-2017. Green area represents the historic range of *Myotis septentrionalis* in Georgia.

2.2 Field Data Collection

We captured bats on state and federally managed lands from mid-May through early-August 2014 – 2017. We followed the Indiana bat survey protocol (U.S. Fish and Wildlife Service 2015), which included mist-netting beginning at sunset and continuing for five hours on nights with no precipitation, no sustained high winds, and temperatures above 10°C. We netted sites for 2-3 consecutive nights predominately using double high mist-nets. We fitted all captured *M. septentrionalis* with radio transmitters (LB- 2X, 0.27 g, Holohil Systems, Ontario, CA)

between the scapulae using surgical adhesive. Transmitter mass did not exceed 5% of body mass (mean body mass = 6.8, range 5.5 - 8 g, Aldridge and Brigham 1988). We released individuals at the capture site and tracked them to diurnal roosts daily until the transmitter fell off or the individual could not be located for 3 consecutive days. When possible, we conducted emergence counts at roosts to confirm roost locations and colony size. Bat capture was conducted under Georgia Scientific Collecting Permit nos. # 29-WJH-15-162, 29-WJH-16-201, and 1000598963, and United States Fish and Wildlife Service native endangered species recovery permit number TE60238B-0. University of Georgia Institutional Animal Care and Use Committee (approval nos. A2014 04-022-Y3-A0, A2017 03-005-Y3-A2) approved bat capture, handling, and transmitter attachment.

We characterized roost trees by measuring height (m), decay stage (Anderson et al. 1979), crown class (Helms 1998), species, DBH, distance (m) from roost to nearest taller tree, and distance from roost to nearest canopy tree (Menzel et al. 2002). We defined canopy trees as those within the co-dominate or dominate crown classes (Helms 1998) and DBH as the diameter of a tree 1.37 m from the highest point of ground at the base. We characterized vegetation within 10-m radius plots surrounding roosts by measuring DBH of woody stems > 2.54 cm and recording number of snags. We acquired additional data for roosts located by private contractors and agency biologists from Georgia Department of Natural Resources (GA DNR). In general, these roosts were located in areas where environmental impact statements were required, or in areas thought to have suitable habitat for threatened or endangered bat species. Data on these roosts and surrounding vegetation varied, but DBH, crown class, and whether the roost was alive or dead was known for all but 2 roosts. We did not use roosts acquired from the GA DNR in plot level analysis due to the incomplete descriptions of surrounding habitat.

2.3 Home Range Selection Model

Because the scale at which *M. septentrionalis* responds to predictor variables was unknown (sensu Chandler and Hepinstall-Cymerman 2016), we examined roost selection at the home range and landscape scales. We defined home range scale as a 67-ha circular buffer around the weighted central mean of all locations for each radio-tagged individual. Home range buffer area was based on mean home range of 23 individuals reported in Owen et al. (2003) and Lacki et al. (2009). For analysis, we combined locations for two individuals because they were captured at the same site and roosted together.

We generated pseudo absences within each home range buffer at a 10:1 ratio using a random point generator in ArcMap 10.2 (ESRI 2014). Pseudo absences represented areas within the home range buffer where a roost were absent. A 10:1 ratio was selected because it is the approximate maximum number of pseudo absences to known locations that does not negatively affect predictive accuracy of a generalized linear model (Barbet-Massin et al. 2012). For all roost trees and pseudo absences, we selected predictor variables related to hypotheses surrounding thermoregulation, including slope (degrees), distance to ridge (m), and aspect (Appendix A). Aspect was derived using the USGS 10 m DEM (U.S. Geological Survey 2017), with east/west and north/south calculated using sine and cosine of aspect in radians, respectively. Distance to ridge was straight-line distance (m) from each point/location to nearest ridge as defined by zero accumulation in a flow accumulation raster derived from the 10 m DEM.

2.4 Landscape Selection Model

To examine landscape-scale roost range selection, we compared area where bat roosts were located (hereafter, roost range) to total available roosting area. We defined roost range as an 11-ha circular buffer centered on the weighted mean centroid of all roost locations for each

individual. Roost range buffer size was equivalent to the mean of 100% minimum convex polygons around groups of roost trees in our data and 100% minimum convex reported in Johnson et al. (2012). We defined total available roosting area as a 17.2-km² circular area centered on the weighted mean of all locations for that individual. Buffer area was based on a 2.35-km radius, which was the farthest movement we recorded between roosts.

We generated random pseudo absences at a 10:1 ratio within the total available roosting area using ArcMap. Pseudo absences represented areas on the landscape where roost ranges were absent. For pseudo absences and known roost ranges, we derived percent deciduous, evergreen, and total forest cover, mean slope, mean east/west aspect, and mean north/south aspect (Appendix A.). Forest cover and mean aspect were calculated using an 11-ha circular moving window and 2011 National Land Cover dataset (U.S. Geological Survey 2014) and 10 m DEM, respectively. Mean aspect was converted to east/west and north/south as described above.

2.5 Analysis

For roost-level analysis, we compared DBH, distance to nearest taller tree, and distance to nearest over story tree between live roosts and snag roosts as well as between roosts and surrounding vegetation. We assessed differences using t-tests and considered differences significant if 95% confidence intervals surrounding mean differences did not include zero. To determine probability of roosting in a snag relative to snag density, we used logistic regression with live or snag roost as the response variable and snag density within plots surrounding roost trees as the independent variable. We transformed logistic regression coefficients to odds ratios to interpret how probability of roost use changes relative to predictor variables. We compared Akaike's Information Criterion (AIC) values of snag density and null models to ascertain if snag density explained use of live trees or snags as roosts, with $\Delta AIC < 2$ between models indicating

there was not overwhelming support for one model over the other. Lastly, to determine if *M. septentrionalis* disproportionately used snags as roosts, we used a chi-squared test to compare number of live trees and snags in plots surrounding roosts to number of live trees and snags used as roosts.

Using logistic regression, we fit landscape and home range scale models relating solar exposure to roost habitat for all individuals and a subset of data containing reproductive females. We pooled reproductive female across reproductive status and included two juvenile bats that roosted in a maternity colony. Due to solar exposure being related to slope position and aspect, we included interactions of the aspect variable and slope in all models at both scales (Kumar et al. 1997). Additionally, some home range models included the additive effect of distance to ridge and some landscape scale models included an additive effect of percent deciduous, evergreen, or total forest cover. We used Pearson's correlations (Pearson's $|r| \leq 0.70$) to ensure no correlated predictor variables were in the same model, but found no correlations.

Models were evaluated using AIC with those having $\Delta AIC < 2$ of the top model considered to have support (Burnham and Anderson 2002). We concluded no support for model estimates in cases where no model outperformed the null model. Supported models were model-averaged based on weights derived from AIC values to obtain a single parsimonious model. We calculated model-averaged predictor variables and their 95% confidence interval, with influential predictor variables defined as those having confidence intervals that did not include zero. We transformed coefficients to odds ratios to express how the likelihood of roost occurrence correlated with predictor variables (Hosmer and Lemeshow 2000).

3. Results

We tracked 16 *M. septentrionalis* for a mean of 4.4 ± 2.7 days (range 1-9 days). Tracked individuals consisted of 1 juvenile female, 1 juvenile male, 2 non-reproductive females, 3 non-reproductive males, 7 pregnant, and 2 lactating females. We included data on 37 roosts, 7 from GA DNR, consisting of 23 snags, 12 live trees, and 2 unreported. Bats roosted in cavities, loose bark, and hollow tree boles. We identified 16 tree species used as roosts, of which 8 species were used more than once (Appendix A). The most commonly selected genus of roost tree was *Pinus* (8 trees) and *Quercus* (7 trees). Of 34 roosts for which roost position was determined, 18 roosts were below the canopy.

Bats switched roost trees every 1.8 ± 1.3 days and no individual stayed in a roost longer than 5 consecutive days. Mean movement distance between roosts was 427.8 ± 620.6 m (range 2.8-2,344.7 m). Reproductive females and juveniles of both sexes roosted in colonies whereas non-reproductive females and adult males always roosted alone. The largest roost observed, located in 2014, contained 31 individuals. Number of individuals in the largest colony detected decreased in every subsequent year (2015 – 19 individuals, 2016 – 3 individuals, and 2017 – 1 individual).

Snags were used as roosts more frequently than expected ($\chi^2 = 66.72$, $P < 0.001$) based on their prevalence. DBH of snag roosts (34.7 ± 24.4 , 20) was greater ($t_1 = 3.71$, $P = 0.001$) than snags in the surrounding area (18.1 ± 6.1 , 20). Likelihood of reproductive females and all individuals roosting in snags was positively correlated to snag density ([reproductive females coefficient] 0.021, 95% confidence interval [CI] = 0.000 - 0.042; [all individuals coefficient] 0.026, CI = 0.005 - 0.048 [Appendix B]). Use of a snag roost was 1.02 and 1.03 times more likely for every 1 snag/ha increase for reproductive females and all individuals, respectively (Fig.

2). Roost tree DBH for reproductive females (31.1 ± 20.9 , 22) and all individuals (29.8 ± 22.0 , 29) was greater ([reproductive females] $t_1 = 3.66$, $P = 0.001$; [all individuals] $t_1 = 3.69$, $P = 0.001$) than mean tree DBH within the surrounding plot ([reproductive females] 14.4 ± 4.2 , 22; [all individuals] 14.7 ± 3.8 , 29). Unless otherwise stated, the format of parenthesis is: mean $\pm$ standard deviation, number of samples.

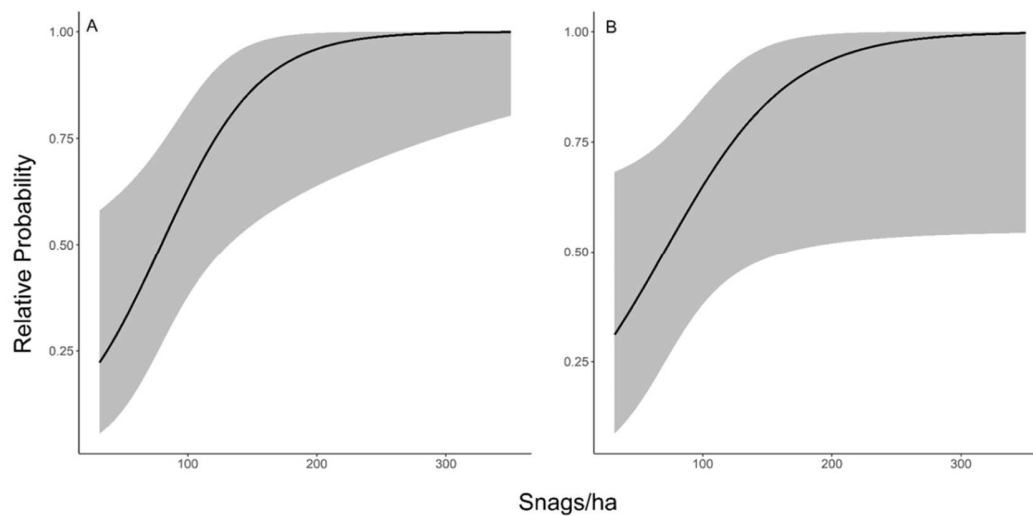


Figure 2. Relative probability (black line) and 95% confidence interval (gray band) that an A) individual or B) reproductive female *Myotis septentrionalis* roost in north Georgia, 2014-2017, was located in a snag based on snag density (snags/ha).

Our most supported models ($\Delta AIC < 2$) for all individuals and reproductive females at the home range scale indicated that effects of slope, aspect, interaction of slope and aspect, and distance to ridge were important in roost tree selection (Table 1). For reproductive females and all individuals, aspect and interaction of slope and aspect were influential (Fig. 2). Reproductive females and all individuals disproportionately roosted on west facing slopes, with individuals being 10.9 and 9.09 times more likely to select a roost at 270° (west) than 90° (east), respectively. Interaction of slope and aspect revealed reproductive females and all individuals

selected west facing aspects on low-grade slopes, but probability diminished as slope increased (Fig. 3). While distance to ridge was in the top model, its effect on roost selection was small and the 95% confidence interval surrounding the estimate included zero. Models evaluating roost range selection for reproductive females and all individuals at the landscape scale failed to outperform the null model and thus were deemed to have no support.

Table 1. Number of parameters (K), Akaike information criterion (AIC) value, difference in AIC from top model (Δ AIC), and AIC weight of all models with Δ AIC < 2 of the top model used to predict roost occurrence of all individuals and reproductive female *Myotis septentrionalis* at the home range scale in north Georgia, USA, 2014-2017. Models for each group (all individuals and reproductive females) were averaged based on AIC weights.

Model	K	AIC	Δ AIC	Weight
All Individuals				
East/West Aspect x Slope	4	246.37	0	0.7
East/West Aspect x Slope + Distance to Ridge	5	248.09	1.71	0.3
Reproductive Females				
East/West Aspect x Slope	4	186.99	0	0.71
East/West Aspect x Slope + Distance to Ridge	5	188.8	1.8	0.29

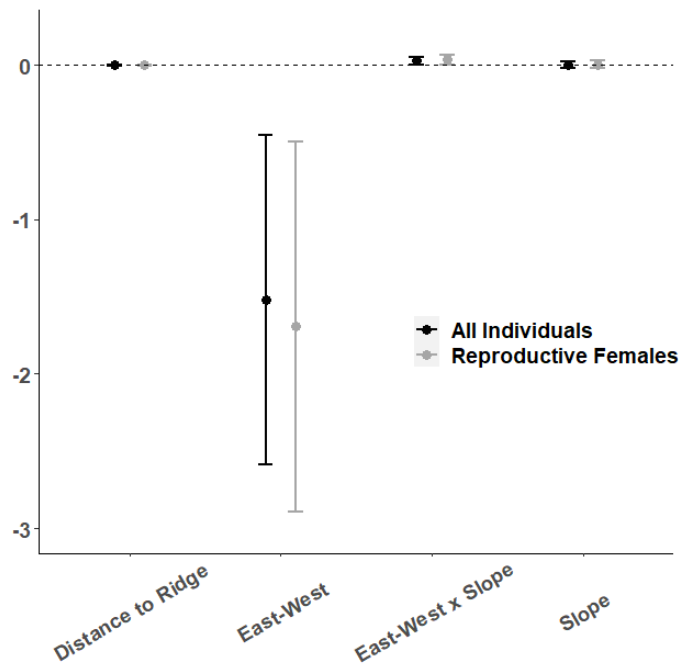


Figure 2. Estimates and 95% confidence intervals of predictor variables used to estimate relative probability of roost occurrence for all individuals and reproductive female *Myotis septentrionalis* at the home range scale in north Georgia, USA, 2014-2017.

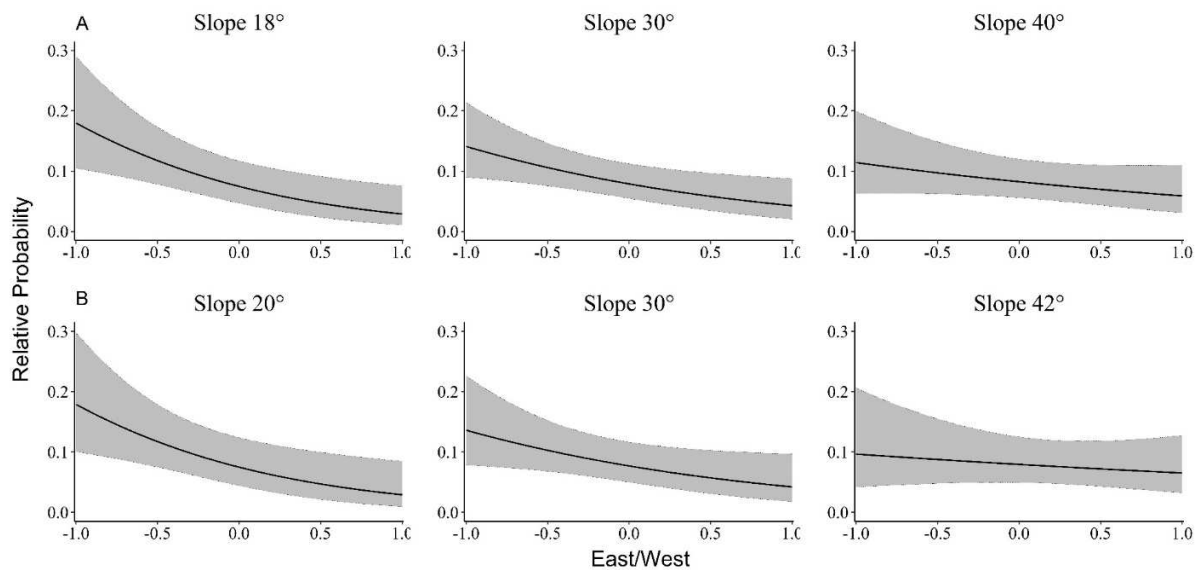


Figure 3. Relative probability of roost occurrence (black line) and 95% confidence intervals (gray bands) at the home range scale for A) all individuals and B) reproductive female *Myotis septentrionalis* in north Georgia, 2014-2017, based on the interactive effect of aspect and slope. Slope value constant above each subplot represents the range of values from 37 roost trees. Constants represent the 25th percentile, mean, and 75th percentile. X-axis values represent sine of aspect in radians, with -1 indicating due west and 1 indicating due east.

4. Discussion

While some studies have documented *M. septentrionalis* preferentially selecting tree species as roosts (Johnson et al. 2009, Silvis et al. 2015a), we documented no apparent preference for any one species. Additionally, while roost tree species preference has been documented, tree of preference varies among study sites (Johnson et al. 2009, Silvis et al. 2015a). Indifference in roost tree species in our study could be explained by our larger study area, which included a wider range of habitats, and consequently tree species, relative to other studies. Roost studies across *M. septentrionalis*' range reveal adaptability in roosting behavior to local conditions, with the species found in pine (Perry and Thill 2007) and hardwood forests (Silvis et al. 2012), and exclusively using Ponderosa pine (*Pinus ponderosa*; Cryan et al. 2001) which occurs in a small portion in the western reaches of the range. Given observed plasticity in tree species used as roosts across the range, selection is likely tied thermoregulatory needs and availability.

Similar to previous research, we found *M. septentrionalis* roosting in live trees and snags, with snags selected more than expected based on availability (Lacki and Schwierjohann 2001, Lacki et al. 2009, Silvis et al. 2015a). While live trees can have cavities and loose bark for roosting, snags are more likely to contain these features, resulting in their disproportionate use as roosts. While DBH was variable, roosts tended to have a larger DBH than surrounding trees and

snags. Larger roosts likely provide a more stable thermal environment, which decreases energetic demands and promotes gestation and development of young (Kunz and Lumsden 2003, Barclay and Kurta 2007, Coombs et al. 2010). Selection for larger snags also may allow for greater roost retention between years, as larger snags often take longer to decay, are less likely to be downed during extreme weather events, and are more prone to develop or retain cavities and loose bark needed for roosting (Vanderwel et al. 2006, Barclay and Kurta 2007, Parish et al. 2010).

While previous studies have examined differences in snag density between roost and random plots (Lacki and Schwierjohann 2001, Lacki et al. 2009, Alston et al. 2019), to our knowledge ours is the first to examine effect of snag density on *M. septentrionalis* roost selection. Propensity of *M. septentrionalis* to select snags relative to their availability, combined with snag use increasing with snag density, demonstrates the importance of snags as roost habitat. However, it should be noted that *M. septentrionalis* have been documented using live trees as roosts even in areas of high snag density, indicating live trees may contain desirable attributes not found in snags (Carter and Feldhamer 2005, Perry and Thill 2007).

Similar to Menzel et al. (2002) and Silvis et al. (2012), we found no relationship between roost selection and increased solar exposure. This finding runs counter to female *M. septentrionalis* at more norther latitudes which roost in areas of high solar exposure, during energetically demanding times (Garroway and Broders 2008, Hyzy et al. 2020). Avoidance of solar exposure was also observed in roost selection relative to canopy height, with over half of the roosts in our study located below the canopy, a selection strategy utilized during warmer years by *M. septentrionalis* in central portions of the range (Silvis et al. 2015a). Reproductive females and all individuals preference for west-facing aspects on low-grade slopes coincides with low solar exposure in the afternoon, while their lack of preference on steeper slopes

indicates an apparent indifference to solar exposure. Additionally, a similar trend in aspect selection between the two groups indicates that the maintenance of higher body temperature in reproductive females, relative to non-reproductive individuals, was achievable under conditions of lower solar exposure (Grinevitch et al. 1995, Johnson and Lacki 2014). If roost selection was based on increased solar exposure, we would expect bats to disproportionally select south-facing or east facing slopes, which receive the most solar exposure and solar exposure during the coolest parts of the day, respectively (Riskin and Pybus 1998, Willis and Brigham 2007, Hammond et al. 2016). While avoidance of solar radiation could indicate selecting for more optimal summer torpor locations (Dzal and Brigham 2013, Johnson and Lacki 2014), the observed variation between regions, similarities in aspect selection between reproductive females and all individuals, and propensity to select roosts below the canopy suggests that the optimal thermal condition is driven by more than daily torpor bouts.

Behavioral adaptation to thermal extremes, like those we documented, highlight the importance of understanding habitat use across a species range and the need to modify management by region. While we found bats in areas thought to reduce roost temperature, there are a number of other factors that can influence roost thermal conditions at the microclimate scale, including ambient temperature, number of individuals present, and location of individuals within the roost structure (Boyles 2007). Due to time restraints and the complexity of measuring roost microclimates we did not attempt to quantify their effects of roost selection. However, factors related to roost microclimate are thought to be less influential than social thermoregulation and ambient conditions and would likely not change the results of our study (Willis and Brigham 2007, Clement and Castleberry 2013).

Similar to other studies, we found *M. septentrionalis* utilized multiple roost trees (Johnson et al. 2012, Silvis et al. 2015b), but our landscape scale analysis did not reveal a relationship between the roost range and surrounding landscape features, indicating that bats do not appear to select roosting habitat at the landscape scale, at least as depicted with the variables investigated. Instead, *M. septentrionalis* could be responding to non-roost related cues, such as large forest patches, when selecting habitat at the landscape scale, then establishing a roost network within the selected area. Alternatively, bats could be selecting areas with diverse landscape features that accommodate all potential needs, thus making the composite of landscape features within the roost range statistically indistinguishable from the surrounding landscape. Lack of relationship between area centers and surrounding landscape could be due to our small sample size (15 roost ranges) for comparison. The limited number may have made discerning patterns of selection more difficult. Additionally, several roost ranges were established from few roosts and may not have accurately reflected the roost range.

5. Conclusion

With *M. septentrionalis* experiencing > 90% mortality from WNS, conservation of quality roost habitat, particularly maternity colonies, is vital to the persistence and recruitment of individuals (Cheng et al. 2021). Our findings demonstrate regional variation in behavior and habitat selection of *M. septentrionalis* and highlight the need for adaptive habitat management across the range. Based on our results, land managers interested in conserving *M. septentrionalis* should focus management on snag creation and retention, with efforts promoting creation of larger diameter snags. In the southern extent of *M. septentrionalis*' range, snag generation should occur across a range of canopy cover classes to encompass varying levels of solar exposures. Additionally, roosting habitat with limited solar exposure may become increasingly important

throughout the range as global temperatures continue to rise (Root et al. 2003). Previous studies have demonstrated benefits of fire to *M. septentrionalis* roost habitat (Johnson et al. 2009, Lacki et al. 2009, Ford et al. 2016), thus prescribed burning, particularly low-intensity burnings that retain the majority of the canopy while killing understory trees, could be used to promote roosting habitat (Van Lear and Harlow 2002, Perry 2012, Harper et al. 2016). We note that our sample size precluded us from investigating variation in roost selection between female reproductive classes. Thus, more information on roosting habits of *M. septentrionalis* in the southern reaches of its range is needed before a clear pattern of thermal requirements can be established.

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Appendix A. Predictor variables used in home range and landscape scale models, their source data, and their mean $\pm$ standard deviation. Home range and landscape scale models designated as female include only reproductive females.

Source		Home Range		Home Range Female		Landscape		Landscape Female	
		Roost	Pseudo-absence	Roost	Pseudo-absence	Roost Range	Pseudo-absence	Roost Range	Pseudo-absence
Aspect East\West	calculated using the sine of the radians from USGS 10 m digital elevation models (DEM)	-0.26 + 0.77	0.03 + 0.75	-0.18 + 0.82	0.08 + 0.74	-0.06 + 0.74	-0.02 + 0.75	0.01 + 0.75	0.05 + 0.74
Aspect North\South	calculated using the cosine of the radians from USGS 10 m digital elevation models (DEM)	0.00 + 0.61	0.12 + 0.65	0.01 + 0.58	0.14 + 0.65	-0.56 + 0.42	-0.45 + 0.5	-0.53 + 0.48	-0.44 + 0.52
Slope	calculated using the USGS 10 m digital elevation models (DEM)	31.26 + 17.36	31.69 + 51.93	32.55 + 18.85	31.69 + 16.76	15.75 + 5.33	18.19 + 6.83	17.6 + 4.75	19.16 + 6.7
Distance to Ridge	Straight-line distance from roost to nearest ridge calculated using the inverse of the USGS 10 m digital elevation models (DEM)	128.58 + 86.44	125.08 + 79.6	128.48 + 87.02	123.63 + 78.42	NA	NA	NA	NA
Percent Forest	Mixed, Deciduous, and Evergreen forest from 2011 National Land Cover Dataset	NA	NA	NA	NA	91.26 + 12.01	93.63 + 17.78	98.62 + 3.33	95.69 + 13.7
Percent Deciduous Forest	Deciduous forest from 2011 National Land Cover Dataset	NA	NA	NA	NA	68.28 + 33.23	77.38 + 29.47	79.66 + 26.58	82.45 + 26.85
Percent Evergreen Forest	Deciduous evergreen forest from 2011 National Land Cover Dataset	NA	NA	NA	NA	17.24 + 25.31	11.66 + 21.07	13.45 + 22.61	9.79 + 20.06

Appendix B. Roost tree species used by *Myotis septentrionalis* in north Georgia, USA, 2014-2017.

Count represents the number of times species were used as a roost.

Common Name	Scientific Name	Count	Live	Dead
Sourwood	<i>Oxydendrum arboreum</i>	4	2	2
Chestnut Oak	<i>Quercus prinus</i>	3	2	1
Loblolly Pine	<i>Pinus taeda</i>	3	1	2
Red Maple	<i>Acer rubrum</i>	3	1	2
Tulip Poplar	<i>Liriodendron tulipifera</i>	3	1	2
White Oak	<i>Quercus alba</i>	3	1	2
White Pine	<i>Pinus strobus</i>	3	0	3
Pignut Hickory	<i>Carya glabra</i>	2	1	1
Black Locust	<i>Robinia pseudoacacia</i>	1	0	1
Black Tupelo	<i>Nyssa sylvatica</i>	1	1	0
Eastern Hophornbeam	<i>Ostrya virginiana</i>	1	1	0
Flowering Dogwood	<i>Cornus florida</i>	1	1	0
Sassafras	<i>Sassafras albidum</i>	1	0	1
Scarlet Oak	<i>Quercus coccinea</i>	1	0	0
Short Leaf Pine	<i>Pinus echinata</i>	1	0	1
Virginia Pine	<i>Pinus virginiana</i>	1	0	1
Hickory sp.	<i>Carya</i> sp.	1	0	1
Unknown	Unknown	4	0	3

Appendix C. Model relating probability of a *Myotis septentrionalis* roost being in a snag based on snag

density and the corresponding null model. Included for each model are the parameter estimate (top

number), 95% confidence interval around the estimate (bottom number), and the Akaike's information

criterion (AIC) value.

Model	Intercept	Snag Density	AIC
All Individuals	-2.087 (-4.236, 0.061)	0.026 (0.005 - 0.048)	26.956
Reproductive Females	-1.457 (-3.553 - 0.64)	0.021 (0.000 - 0.042)	24.02
Null	0.747 (-0.046, 1.540)		37.165